

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN DRAINAGE SYSTEM

MODULE - 3

**PENINSULAR & HIMALAYAN
RIVER SYSTEM**

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

2

THE GANGA SYSTEM

The Ganga is the most important river of India both from the point of view of its basin and cultural significance.

ORIGIN OF THE RIVER

HEADSTREAM OF THE RIVER

1

It rises in the **Gangotri glacier** near Gaumukh (3,900 m) in the **Uttarkashi district of Uttaranchal**.

2

Here, it is known as the **Bhagirathi**. It cuts through the Central and the Lesser Himalayas in narrow gorges.

3

At **Devprayag**, the Bhagirathi meets the **Alaknanda**; hereafter, it is known as the Ganga.

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

2 THE GANGA SYSTEM



4 The Alaknanda has its source in the **Satopanth glacier** above **Badrinath**.

5 The Alaknanda consists of the **Dhaul** and the Vishnu Ganga which meet at Joshimath or **Vishnu Prayag**.

6 The other tributaries of Alaknanda such as the Pindar join it at **Karna Prayag** while Mandakini meets it at **Rudra Prayag**.

7 The Ganga enters the plains at Haridwar.

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

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THE GANGA SYSTEM



*An Aerial view of the Hooghly River
viewed over the town of Howrah*

8 From here, it flows first to the south, then to the south-east and east and flows as **the distributary of Hooghly-Bhagirathi** and the entire river system has a total length of 2,525 km.

9 It flows through the states of Uttarakhand (110 km), Uttar Pradesh (1,450 km), Bihar (445 km) and West Bengal (520 km).

10 The Ganga basin covers about 8.6 lakh sq. km area in India alone.



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THE GANGA SYSTEM

11

- The Ganga river system is the largest in India having a number of perennial and non-perennial rivers originating in the Himalayas in the north and the Peninsula in the south, respectively.
- The river finally discharges itself into the Bay of Bengal **near the Sagar Island.**

RIGHT BANK TRIBUTARY

The important Right bank tributaries are:

- Yamuna
- Son

LEFT BANK TRIBUTARY.

The important left bank tributaries are:

- The Ramganga
- the Gomati
- the Ghaghra
- the Gandak
- the Kosi and
- the Mahananda.

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THE GANGA SYSTEM

YAMUNA RIVER

12

- The Yamuna is the western most and the longest tributary of the Ganga.
- It has its source in the Yamunotri glacier on the western slopes of Banderpunch range (6,316 km).
- It joins the Ganga at Triveni Sangam/Prayagraj (Allahabad).

RIGHT BANK TRIBUTARY OF YAMUNA

- the Chambal
- the Sind
- the Betwa
- the Ken
- the Tons

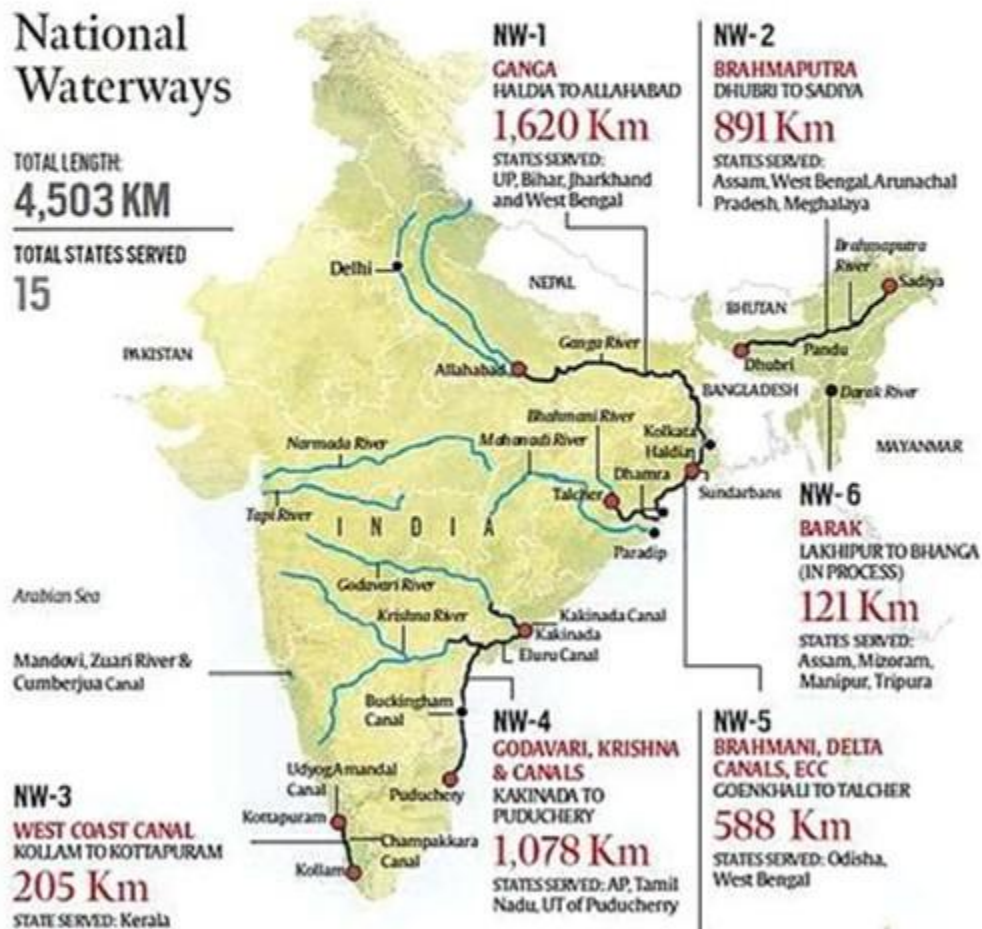
LEFT BANK TRIBUTARY OF YAMUNA

- the Hindon
- the Rind
- the Sengar
- the Varuna, etc.

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

QUICK FACTS FOR EXAMS!

TOTAL LENGTH:
4,503 KM

TOTAL STATES SERVED
15

IMPORTANT NATIONAL WATER WAYS OF THE HIMALAYAN DRAINAGE

NAME	RIVER
NW1	Prayagraj-Haldia stretch of the Ganga-Bhagirathi-Hooghly Rivers
NW2	Sadiya-Dhubri stretch of Brahmaputra River
NW110	Yamuna River

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

QUICK FACTS FOR EXAMS!



DID YOU KNOW?

- The Ganga River is known by different names when it gets joined by different rivers.
- Near Rajmahal Hills it turns to the south of Farraka, it ceases to be known as Ganga.
- It **bifurcates itself into Bhagirathi-Hugli** in West Bengal.
- It flows as **Padma** into Bangladesh where it meets **Meghna** River.
- Likewise, the Brahmaputra upon reaching Bangladesh **meets Teesta and known as Jamuna**.
- The Jamuna River eventually meets the Padma River.

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THE GANGA SYSTEM

CHAMBAL



- 13** The Chambal is famous for its **badland topography** called the **Chambal ravines**.

- 14** The Chambal rises near **Mhow** in the **Malwa plateau of Madhya Pradesh** and flows northwards through a gorge upwards of **Kota** in Rajasthan, where the **Gandhisagar dam** has been constructed

- 15** From **Kota**, it traverses down to **Bundi**, **Sawai Madhopur** and **Dholpur**, and finally joins the **Yamuna**.

- 16** The **Banas** is an important tributary of the Chambal. It originates in the southern part of the **Aravali Ranges** and joins the **Chambal near Sawai Madhopur**

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

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THE GANGA SYSTEM

QUICK FACTS ABOUT CHAMBAL

17 National Chambal Gharial Sanctuary:

- This is a **tristate project** to facilitate the restoration of ecological health of the river system and to provide full protection for the **gharial** species.
- The 3 states under which the project falls are **Madhya Pradesh, Uttar Pradesh, Rajasthan**.

18 Dams on the Chambal River:

Given below are the names of the important dams on the Chambal River which is used for **Hydro**

Power Generation

- **Ranapratap Sagar Dam**
- **Gandhi Sagar Dam**
- **Jawahar Sagar Dam**
- **Kota Barrage.**

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THE GANGA SYSTEM

GANDAK

- 19 The Gandak comprises two streams, namely **Kaligandak** and **Trishulganga**.
- 20 It rises in the Nepal Himalayas between the Dhaulagiri and Mount Everest and drains the central part of Nepal.
- 21 It enters the Ganga plain in Champaran district of Bihar and joins the Ganga at Sonpur near Patna.
- 22 A large number of its tributaries rises in Nepal Himalayas, the most outstanding being the **Kali Gandak**.

GHAGHRA

- 23 It is also known as Karnaili in Western Nepal.
- 24 Its important tributaries are the **Sarda**, the Sarju on the bank of which Ayodhya is located and the Rapti.
- 25 The Ghaghara originates in the glaciers of Mapchachungo in Tibet.

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THE GANGA SYSTEM

SARDA OR KALI GANGA

- 26** The river **Sarda** (Mahakali or Kali or **Kali Ganga**) joins Ghaghra in the plain before it finally meets the Ganga at Chhapra.



Sagarmatha National Park

KOSI

- 27** The **Kosi river** is a **trans boundary river** between India and Nepal. It consists of 7 streams and it is thus known as **Sapt Kosi**.
- 28** The Kosi is an antecedent river with its source to the north of Mount Everest in Tibet, where its **main stream Arun rises**.
- 29** After crossing the Central Himalayas in Nepal, it is joined by the Son Kosi from the West and the Tamur Kosi from the east.
- 30** **Sagarmatha National Park** is present in the upper catchment areas of Kosi in Nepal.

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THE GANGA SYSTEM



Ramganga river

RAMGANGA

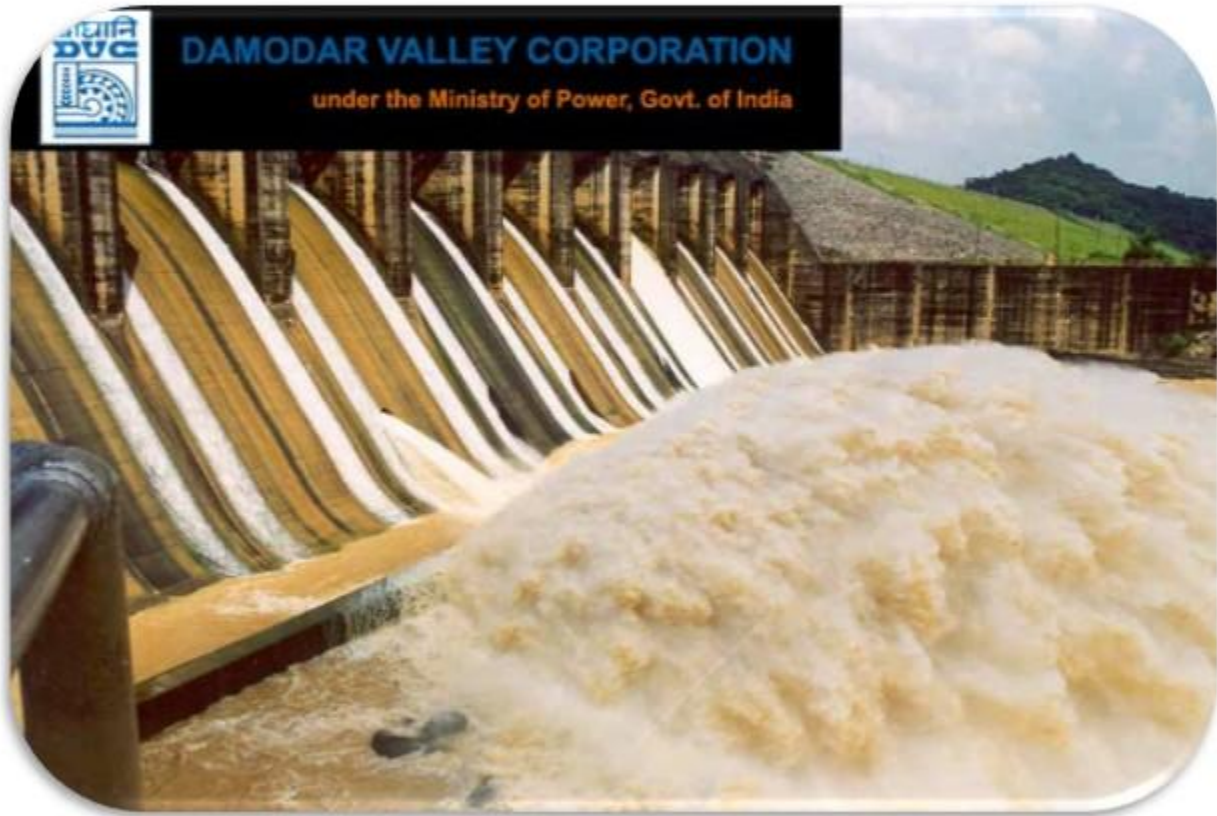
- 31 The Ramganga is comparatively a small river rising in the **Garhwal hills**.
- 32 It changes its course to the southwest direction after crossing the Shiwalik and enters into the plains of Uttar Pradesh near Najibabad.
- 33 Finally, it joins the Ganga near **Kannauj**.

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THE GANGA SYSTEM



Damodar Valley Corporation

DAMODAR

- 34 The Damodar occupies the **eastern margins of the Chotanagpur Plateau** where it flows through a rift valley and finally joins the **Hugli**.
- 35 The **Barakar** is its main tributary.
- 36 Once known as the 'Sorrow of Bengal', the Damodar has been now tamed by the Damodar Valley Corporation, a multipurpose project.

INDIAN DRAINAGE SYSTEM

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THE GANGA SYSTEM

MAHANANDA

37 The Mahananda is another important tributary of the Ganga rising in the **Darjiling hills**.

38 It joins the Ganga as its last left bank tributary in West Bengal.



R.Mahananda

SON

39 The Son is a large south bank tributary of the Ganga, originating in the **Amarkantak plateau**.

40 The important tributaries of the Son are the Johilla, the Gopat, the **Rihand**, the Kanhar and the **North Koel**.



Son river

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

3 THE BRAHMAPUTRA SYSTEM

ORIGIN OF THE RIVER

DIFFERENT NAMES OF THE RIVER



1 The Brahmaputra, one of the largest rivers of the world, has its origin in the **Chemayungdung glacier** of the Kailash range near the Mansarovar lake.

- 2**
- Different names of Brahmaputra: **Yarlung Tsangpo, Dihang, Brahmaputra.**
 - From here, it traverses eastward longitudinally for a distance of nearly 1,200 km in a dry and flat region of southern Tibet, where it is known as the **Tsangpo, which means 'the purifier.'**

INDIAN DRAINAGE SYSTEM

THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

3 THE BRAHMAPUTRA SYSTEM

- 3 The **Rango Tsangpo** is the major right bank tributary of this river in Tibet.



The Brahmaputra makes a deep U-turn near Namcha Barwa

- 4 It emerges as a turbulent and dynamic river after carving out a deep gorge in the Central Himalayas near **Namcha Barwa** (7,755 m).

- 5 The river emerges from the foothills under the name of **Siang or Dihang**.

- 6 It enters India west of **Sadiya** town in Arunachal Pradesh.

- 7 Flowing southwest, it receives its main left bank tributaries, viz., **Dibang or Sikang and Lohit**; thereafter, it is known as the Brahmaputra.

BRAHMAPUTRA RIVER

Origin: Himalayas of Tibet, CHINA



THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

3 THE BRAHMAPUTRA SYSTEM

- 8 The Brahmaputra receives numerous tributaries in its 750 km long journey through the Assam valley.
- 9 Its major left bank tributaries are the Burhi Dihing, Dhansari (South) and Kalang whereas the important right bank tributaries are the Subansiri, Kameng, Manas and Sankosh.
- 10 The Subansiri which has its origin in Tibet, is an antecedent river.

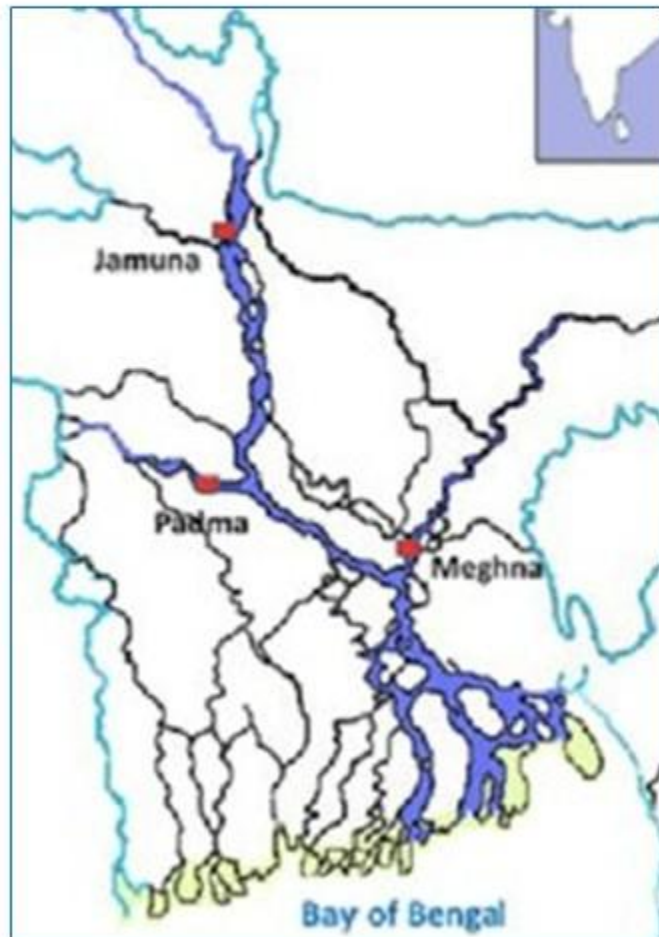
11 Brahmaputra enters India west of Sadiya town in Arunachal Pradesh.

12 The Brahmaputra enters into Bangladesh near Dhubri and flows southward.



THE RIVER SYSTEMS OF THE HIMALAYAN DRAINAGE

3 THE BRAHMAPUTRA SYSTEM



13 In Bangladesh, the Tista joins it on its right bank from where the river is known as the **Jamuna**.

14 It finally **merges with the river Padma**, which falls in the Bay of Bengal.

15 The Brahmaputra is well-known for **floods, channel shifting and bank erosion**.

16 This is due to the fact that most of its tributaries are large, and bring large quantity of sediments owing to heavy rainfall in its catchment area.

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